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Characterization of Message Lengths for Capacity-Achieving Linear PIR

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Main theorem of this study :

$$\{L \in \mathbb{Z}_{\geq 1} \mid \exists \text{ a CALPIR scheme with message length } L\} = \{a(N-1) + bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\}$$

$\mathbb{Z}_{\geq 1} := \mathbb{Z} \cap [1, \infty) = \{1, 2, 3, \dots\}$

What is PIR (in Information Theory) ?

- Private Information Retrieval (PIR) allows a user to retrieve one of $M \in \mathbb{Z}_{\geq 2}$ messages replicated across $N \in \mathbb{Z}_{\geq 2}$ databases (DBs), without revealing the desired message index $\theta \in [M] := \{1, 2, \dots, M\}$ to any DB.
- The (inverse of) PIR capacity, the minimum ratio of expected download size to message length $L \in \mathbb{Z}_{\geq 1}$ is $\dfrac{1 - \frac{1}{N^M}}{1 - \frac{1}{N}}$.

Previous studies

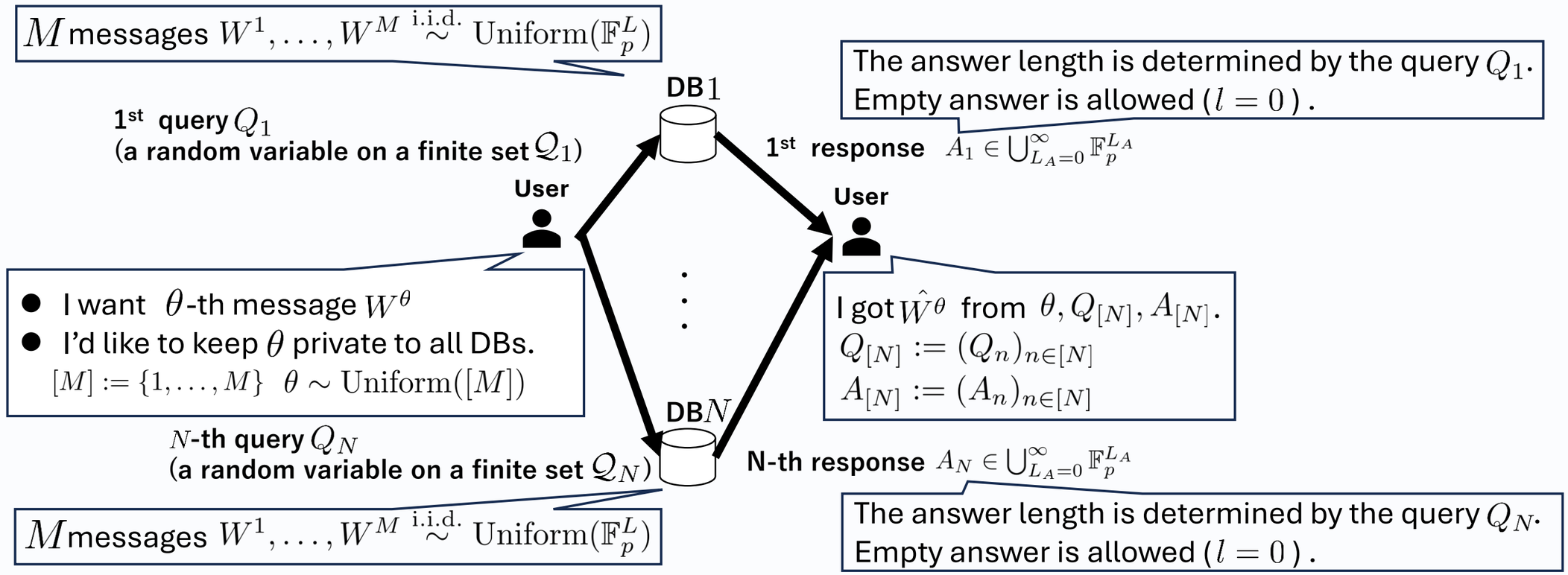
- Previous constructions of Capacity-Achieving Linear PIR (CALPIR) schemes achieve message lengths with
$$L = N - 1 \text{ and } L = N^{M-1}$$
Combining these, we can construct a CALPIR scheme with
$$L \in \{a(N-1) + bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\}$$
- It is known that minimum message length of a CALPIR is $L = N - 1$.

To understand the structure of CALPIR, we characterize all achievable message lengths.

Our main result : We show the converse. If a CALPIR exists, then $L \in \{a(N-1) + bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\}$.
Therefore, **a CALPIR scheme with message length** $L \in \mathbb{Z}_{\geq 1}$ **exists if and only if**

$$L \in \{a(N-1) + bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\}$$

System model of PIR



Parameters : $M, N \in \mathbb{Z}_{\geq 2}, L \in \mathbb{Z}_{\geq 1},\ p$: a prime power

Conditions :

- Query independency : $\mathrm{I}(\theta, Q_{[N]}\ ;\ W^{[M]}) = 0$ $W^{[M]} := (W^m)_{m\in[M]}$
- Answer functionality : $\forall n,\ \mathrm{H}(A_n | Q_n, W^{[M]}) = 0$
- Non-colluding, message replicated databases (DBs).
- Desired message correctness : it always holds that $\hat{W}^\theta = W^\theta$, i.e. $\forall m, \mathrm{H}(W^m | Q_N, A_{[N]}, \theta = m) = 0$
- Index privacy : $\forall n, m, m', q,\ \Pr(Q_n = q | \theta = m) = \Pr(Q_n = q | \theta = m')$

Performance evaluation criteria :

- Normalized download cost

$$\frac{\sum_{n\in[N]} \mathrm{H}(A_n | Q_n)}{L}$$

  - By index privacy, the download cost is independent of m.
  - Let base of the logarithm be the order of the finite field p.
  - For linear PIR, this cost is equivalent to that used in [Sun+, 2017 (1)] and [Tian+, 2019].

- We ignore upload cost.

Previous studies on constructions of CALPIR schemes

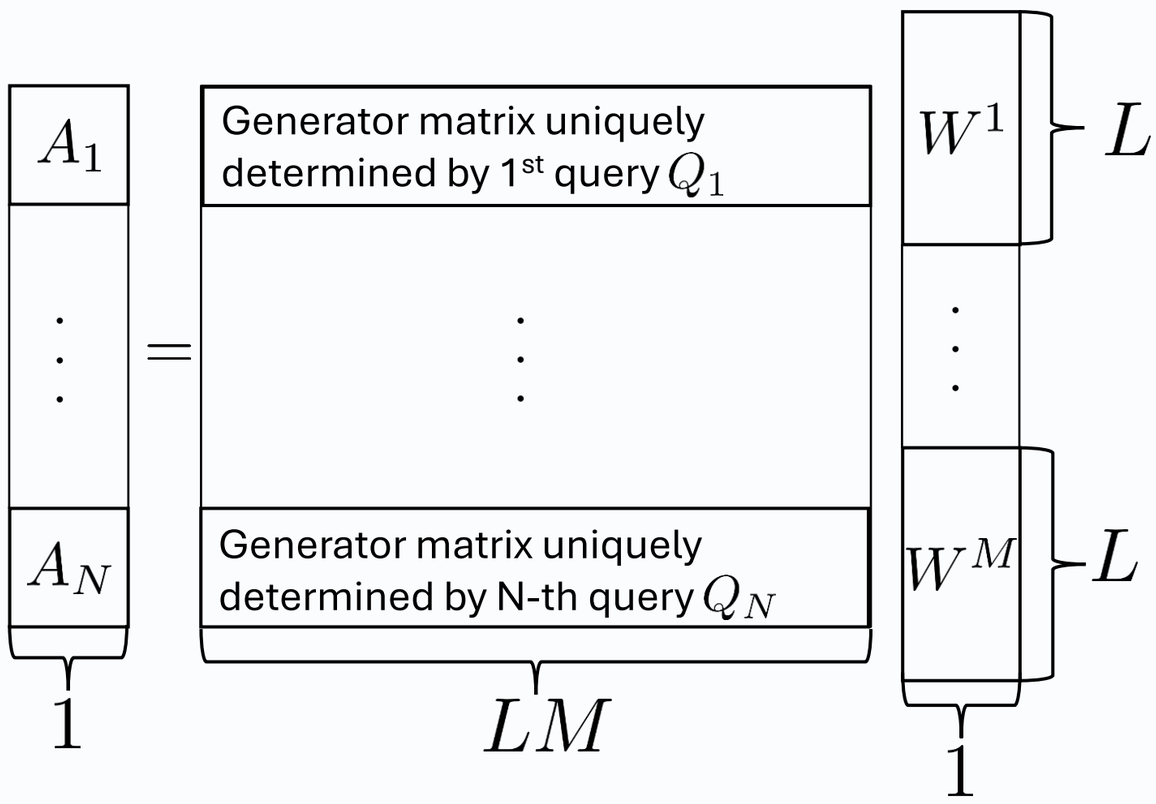
Proposition ((inverse of) PIR capacity) [Sun +, 2017 (1)]

$$\forall L, M, N, p, \quad \frac{\sum_{n\in[N]} \mathrm{H}(A_n|Q_n)}{L} \geq \frac{1-\frac{1}{N^M}}{1-\frac{1}{N}}$$

Previous construction of CALPIR

- [Sun +, 2017 (2)] constructed a CALPIR scheme with message length $L = N^{M-1}$ (over any finite field $\mathbb{F}_p$).
- [Tian +, 2019] constructed a CALPIR scheme with message length $L = N - 1$ (over any finite field $\mathbb{F}_p$).
  [Tian +, 2019] also shows $L = N - 1$ is the minimum achievable.

Definition (Linear PIR)



This study : characterization of message length of CALPIR

We characterize all message lengths L of CALPIRs. To our knowledge, no previous study has characterized L completely.

- There exists a CALPIR scheme for every message length $L \in \{a(N-1)+bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\}$.
  by taking the direct sum of a (overall) generator matrices for [Tian+, 2019] and b (overall) generator matrices for [Sun+, 2017 (2)].
- **This study : (Converse part) No other positive message lengths are achievable.**

Main theorem of this study (characterization of message length for CALPIR schemes)
Let $\mathbb{F}_p$ be any finite field. Let $M, N \in \mathbb{Z}_{\geq 2}$. Then, a CALPIR scheme with message length $L \in \mathbb{Z}_{\geq 1}$ exists if and only if

$$L \in \{a(N-1)+bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\}$$

Proof of main theorem (1/3)
By previous results, it remains to prove the converse : if a CALPIR exists, then $L \in \{a(N-1)+bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\}$.
Assume that

$$L\frac{1-\frac{1}{N^M}}{1-\frac{1}{N}} - L \leq \left\lfloor \frac{L}{N-1} \right\rfloor . \text{ : (1)}$$

We set $a := \frac{L-N^{M-1}b}{N-1}$, $b := L - \left\lfloor \frac{L}{N-1} \right\rfloor (N-1)$.

By elementary algebraic calculation, it holds that $a, b \in \mathbb{Z}_{\geq 0}$, $L = a(N-1)+bN^{M-1}$.
It suffices to prove the inequality **(1)**.

Hereafter, we fix
- $m \in [M]$
- $\boldsymbol{q} \in \{\boldsymbol{q}' \in \prod_n \mathcal{Q}_n | \Pr(Q_{[N]} = \boldsymbol{q}' | \theta = m) > 0\}$

Lemma (P1 and P2 in [Tian +, 2019]. We omit the proof.**)**
For any $n \in [N]$,
P1 $\mathrm{H}\left(A_{[N]} | Q_{[N]} = \boldsymbol{q}, \theta = m\right) = \sum_{n \in [N]} \mathrm{H}\left(A_n | Q_{[N]} = \boldsymbol{q}, \theta = m\right)$
P2 $\mathrm{H}\left(A_{[N]} | Q_{[N]} = \boldsymbol{q}, W^m, \theta = m\right) = \mathrm{H}\left(A_n | Q_{[N]} = \boldsymbol{q}, W^m, \theta = m\right)$

By this Lemma, the following inequality and equality hold.

This study : characterization of message length of CALPIR

Proof of main theorem (2/3)

$$
\begin{aligned}
& N\mathrm{H}\left(A_{[N]}\middle|Q_{[N]}=\boldsymbol{q},W^m,\theta=m\right) \\
&= \sum_{n\in[N]} \mathrm{H}\left(A_{[N]}\middle|Q_{[N]}=\boldsymbol{q},W^m,\theta=m\right) && \because \text{The terms of the sum do not depend on } n. \\
&= \sum_{n\in[N]} \mathrm{H}\left(A_n\middle|Q_{[N]}=\boldsymbol{q},W^m,\theta=m\right) && \because \textbf{P2} \\
&\le \sum_{n\in[N]} \mathrm{H}\left(A_n\middle|Q_{[N]}=\boldsymbol{q},\theta=m\right) && \because \textbf{P1} \\
&= \mathrm{H}\left(A_{[N]}\middle|Q_{[N]}=\boldsymbol{q},\theta=m\right) \\
&= \underline{\mathrm{H}\left(W^m,A_{[N]}\middle|Q_{[N]}=\boldsymbol{q},\theta=m\right)} && \because W^m \text{ can be recovered from } A_{[N]},Q_{[N]},\theta \\
&= L+\mathrm{H}\left(A_{[N]}\middle|Q_{[N]}=\boldsymbol{q},W^m,\theta=m\right). && \because \mathrm{H}(W^m)=L,\ \mathrm{I}(\theta,Q_{[N]}\ ;\ W^{[M]})=0
\end{aligned}
$$

Thus, $\underline{\underline{\mathrm{H}\left(A_{[N]}\middle|Q_{[N]}=\boldsymbol{q},W^m,\theta=m\right)}} \le \dfrac{L}{N-1}$: **(2)**

Since

By linearity of CALPIR

- the tuple (W^m, A_N) is **linear function** of $W^{[M]}$ when $Q_{[N]}$ and θ are determined, and
- $W^{[M]}$ follows uniform distribution on $\mathbb{F}_p^{LM}$,

$$\underline{\mathrm{H}\left(W^m,A_{[N]}\middle|Q_{[N]}=\boldsymbol{q},\theta=m\right)} = (\text{rank of the linear map from } W^{[M]} \text{ to } (W^m,A_{[N]})).$$

Thus, $\underline{\mathrm{H}\left(A_{[N]}\middle|Q_{[N]}=\boldsymbol{q},W^m,\theta=m\right)} \in \mathbb{Z}_{\ge 0}$. Thus, by the inequality (2), $\underline{\mathrm{H}\left(A_{[N]}\middle|Q_{[N]}=\boldsymbol{q},W^m,\theta=m\right)} \le \left\lfloor \frac{L}{N-1} \right\rfloor$.

Apply $\sum_{\boldsymbol{q}\in\{\boldsymbol{q}'\in\prod_n \mathcal{Q}_n \mid \Pr(Q_{[N]}=\boldsymbol{q}'\mid\theta=m)>0\}} \Pr\left(Q_{[N]}=\boldsymbol{q}\middle|\theta=m\right)$ to both sides, $\underset{\sim}{\mathrm{H}\left(A_{[N]}\middle|Q_{[N]},W^m,\theta=m\right)} \le \left\lfloor \frac{L}{N-1} \right\rfloor$.

This study : characterization of message length of CALPIR

Proof of main theorem (3/3)

Thus, the inequality (1) holds since

$$
\begin{aligned}
& L\frac{1-\frac{1}{N^M}}{1-\frac{1}{N}} - L \\
&= \sum_{n\in[N]} \mathrm{H}(A_n|Q_n) - L && \because \text{capacity achieving} \\
&= \sum_{n\in[N]} \mathrm{H}(A_n|Q_n,\theta=m) - H(W^m) && \because \text{index privacy, } \mathrm{H}(W^m) = L \\
&= \sum_{n\in[N]} \mathrm{H}(A_n|Q_{[N]},\theta=m) - H(W^m|Q_{[N]},\theta=m) && \because A_n \text{ is determined by } Q_n, W^{[M]},\ \mathrm{I}(\theta, Q_{[N]}\ ;\ W^{[M]}) = 0 \\
&= \mathrm{H}(A_{[N]}|Q_{[N]},\theta=m) - H(W^m|Q_{[N]},\theta=m) && \because \text{Apply } \sum_{\boldsymbol{q}\in\{\boldsymbol{q}'\in\prod_n \mathcal{Q}_n|\Pr(Q_{[N]}=\boldsymbol{q}'|\theta=m)>0\}} \Pr\left(Q_{[N]}=\boldsymbol{q}|\theta=m\right) \text{ to } \mathbf{P1} \\
&= \mathrm{H}(W^m, A_{[N]}|Q_{[N]},\theta=m) - H(W^m|Q_{[N]},\theta=m) \\
&= \underbrace{\mathrm{H}(A_{[N]}|Q_{[N]}, W^m, \theta=m)} \le \left\lfloor \frac{L}{N-1} \right\rfloor . \quad \square
\end{aligned}
$$

Discussion

Novelty and effectiveness of this study

- Novelty : To our knowledge, the complete characterization of achievable message lengths for CALPIR has not been established.
- Effectiveness : Knowing the minimum achievable message length ([Tian +, 2019]) does not determine all achievable lengths.
  Our result gives an exact criterion for capacity achievability at any given message length.

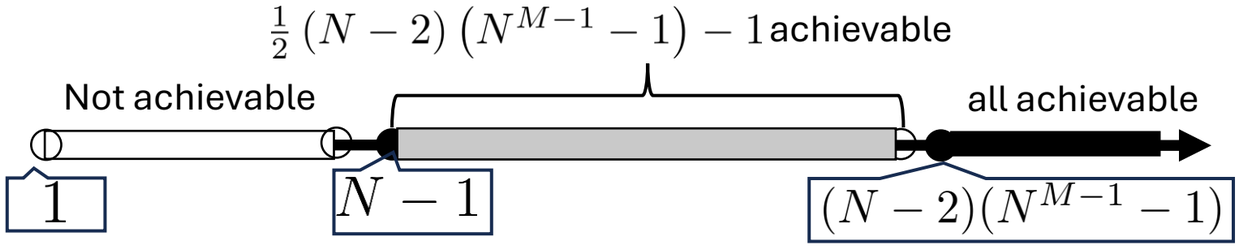
Range of message length of CALPIR schemes

If $N = 2$, then $\{a(N-1) + bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\} = \mathbb{Z}_{\geq 1}$. Thus, a CALPIR exists with every message length.

Assume $N \geq 3$

Since $N-1$ and N^{M-1} are coprime and $N - 1 < N^{M-1}$, the following results hold by elementary number theory.

1. If $L \geq (N-2)(N^{M-1}-1)$, then $L \in \{a(N-1) + bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\}$.
2. If $1 \leq L \leq N-2$, then $L \notin \{a(N-1) + bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\}$.
3. If $N-1 \leq L \leq (N-2)(N^{M-1}-1) - 1$, both cases ($L \in \ldots$ and $L \notin \ldots$) are possible.
   - $(N-2)(N^{M-1}-1) - 1 \notin \{a(N-1) + bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\}$
   - $N - 1 \in \{a(N-1) + bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\}$
   - $\left|\{a(N-1) + bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \cap [N-1, (N-2)\left(N^{M-1}-1\right) - 1]\right| = \frac{1}{2}(N-2)\left(N^{M-1}-1\right) - 1$



Range of message length of CALPIR schemes. The black region is achievable. The white region is not achievable. The gray region contains both achievable and unachievable lengths (when $(M,N) = (2,3)$, this region is empty).

Example: $N = 4, M = 16 \Rightarrow$
$\left|\{a(N-1) + bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \cap [N-1, (N-2)\left(N^{M-1}-1\right) - 1]\right| = 1073741822$
$(N-2)\left(N^{M-1}-1\right) = 2(4^{15}-1) = 2147483646$

Conclusion, Future work, References

Conclusion : A CALPIR scheme with message length $L \in \mathbb{Z}_{\geq 1}$ exists if and only if

$$L \in \{a(N-1) + bN^{M-1} \mid (a,b) \in \mathbb{Z}_{\geq 0} \times \mathbb{Z}_{\geq 0}\} \setminus \{0\}$$

Future work :
Characterization of message length of CALPIR schemes in other models. For example, PIR with MDS coded DBs.

References

- **[Sun +, 2017 (1)]**
  H. Sun and S. A. Jafar, ``The Capacity of Private Information Retrieval,'' IEEE Trans. Inf. Theory, 63(7), 4075–4088, 2017.
- **[Sun +, 2017 (2)]**
  H. Sun and S. A. Jafar, ``Optimal Download Cost of Private Information Retrieval for Arbitrary Message Length,'' IEEE Trans. Inf. Forensics Security, 12(12), 2920–2932, 2017.
- **[Tian +, 2019]**
  C. Tian, H. Sun, and J. Chen,
  ``Capacity-Achieving Private Information Retrieval Codes With Optimal Message Size and Upload Cost,'' IEEE Trans. Inf. Theory, 65(11), 7613–7627, 2019.